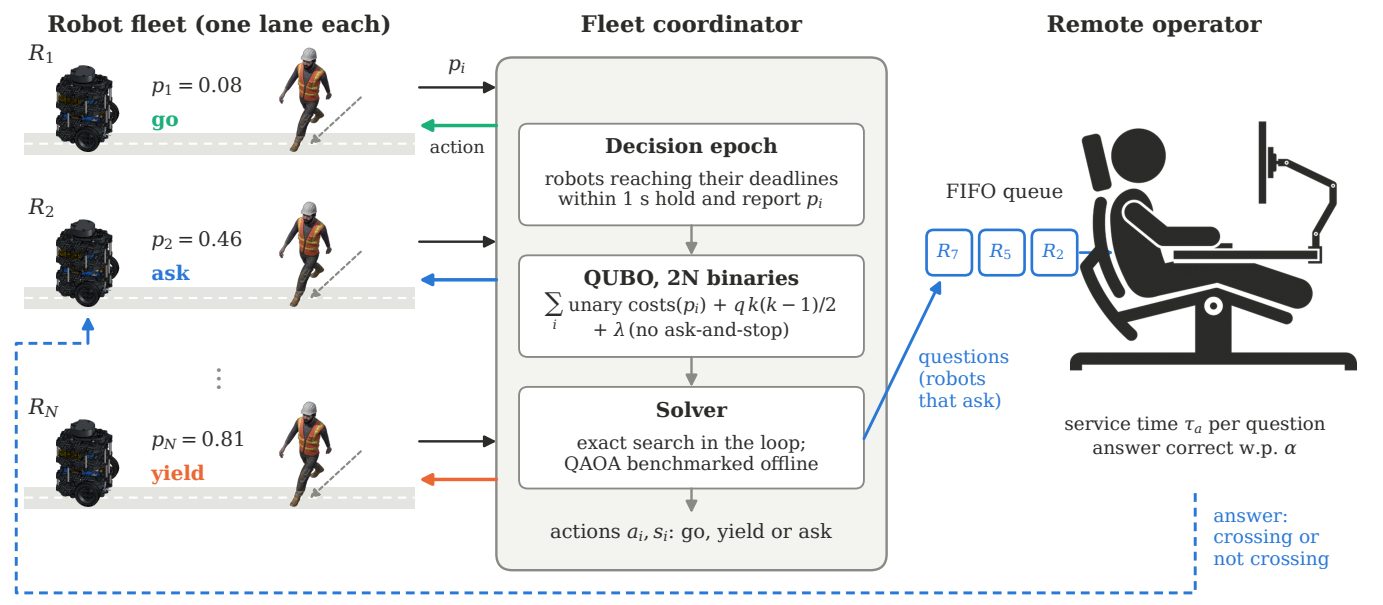

Robot fleet (one lane each)
Fleet coordinator
Remote operator
R_1
$p_1 = 0.08$
go
R_2
$p_2 = 0.46$
ask
R_N
$p_N = 0.81$
yield
p_i
action
Decision epoch
robots reaching their deadlines within 1 s hold and report p_i
QUBO, 2N binaries
$\sum_i$ unary costs(p_i) + $q\,k(k-1)/2$ + λ (no ask-and-stop)
Solver
exact search in the loop; QAOA benchmarked offline
actions a_i, s_i: go, yield or ask
FIFO queue
R_7
R_5
R_2
questions (robots that ask)
service time τ_a per question
answer correct w.p. α
answer: crossing or not crossing